# SELAH FREDERICKSON

## CONTACT


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## EDUCATION

**University of Montana**
Missoula, MT
*Bachelor of Science in Computer Science (Algorithm Design)*
*Bachelor of Arts in Sociology*
*Minors in Mathematics & English*
**Expected Graduation:** May 2027

- Davidson Honors College
- GPA: 4.0

## COURSEWORK

**Completed:**

- CSCI 332 Advanced Data Structures and Algorithms
- CSCI 340 Database Design
- STAT 342 Probability & Simulation
- CSCI 361 Computer Architecture
- CSCI 394 Teaching Computer Science
- SOCI 318 Sociological Research Methods
- DVA 513 *(graduate-level)* Applied Artificial Intelligence

**In Progress:**

- M 361 Discrete Optimization
- M 461 Data Science Analytics
- SOCI 563 *(graduate-level)* Social Data Analysis

## RESEARCH EXPERIENCE

### Research Assistant — Dr. Jacob Downs

University of Montana: September 2025 - Present
**Wildfire History Atlas:**
*Independently developed under faculty mentorship*

- Conducted interdisciplinary research on historical wildfire occurrence using the Monitoring Trends in Burn Severity (MTBS) dataset, combining geospatial analysis, data visualization, and exploratory statistical analysis.
- Developed and expanded an interactive wildfire visualization system from exploratory Python-based visualizations into a modular Dash application incorporating spatial, temporal, environmental, and quantitative analysis.
- Implemented multiple choropleth visualization methods and interactive temporal visualizations to support comparison of wildfire activity across geographic regions and over time.
- Developed a Satellite Mode using Dash Leaflet to integrate wildfire visualizations with satellite imagery and additional basemaps.
- Designed quantitative summaries and statistical analyses to provide additional context for wildfire activity in a selected subregion.

### Research Project — Mälardalen University

Västerås, Sweden: Spring 2026
**A Weighted Approach to Student Persona Modeling in AI-Driven Educational Simulations:**
*Collaborative research project; five-student research team*

- Conducted a literature review of approaches to student persona modeling for AI-driven educational simulations.
- Developed student personas based on characteristics and modeling methods identified in the literature.
- Evaluated persona realism using feedback from practicing teachers and school counselors.

### Research Project — Dr. James Tuttle

University of Montana: August 2026 – Present
**Factors Associated with High School Students' Educational Expectations:**
*Independently developed under faculty mentorship*

- Conducting quantitative research of factors associated with high school students' educational expectations, including mathematics achievement and mathematics self-efficacy.
- Designed the research methodology to analyze a categorical dependent variable using multinomial logistic regression.

## RESEARCH PRESENTATIONS

**Frederickson, S.** *Wildfire History Atlas.* Poster presented at the Smart Fires Conference, University of Montana, September 2026.
**Frederickson, S.** *Wildfire History Atlas*. Research presentation to the Smart Fires AI/ML Team, University of Montana, October 2026.

## SCHOLARSHIPS

- Marie H. Wallace Presidential Scholarship
  2023 – 2027
- UMAA Scholarship
  2026
- Touch America Scholarship
  2025
- Henry and Fannie Clift Kain and Daniel and Jessie McKay Scholarship for Sociology
  2025
- National Merit Finalist Scholarship
  2023
- President's 4.0 GPA List
  2023 - Present

## SERVICE

**Student Evaluation Committee**
Computer Science Department: 2026 - Present
*Faculty-nominated representative*

- Participated in student-led evaluation of Computer Science faculty.
- Consolidated feedback using course evaluations.

**Learning Assistant**
Computers, Ethics, and Society: Spring 2025

- Facilitated small group discussions and provided feedback on student writing.
- Supported the instructor in evaluating active-learning strategies and student engagement through a survey and regular updates.
- Contributed a sociological perspective to an interdisciplinary computing course.

## RESEARCH INTERESTS

- Data visualization; geospatial data analysis
- Human-computer interaction; artificial intelligence; educational technology

## RESEARCH & TECHNICAL SKILLS

**Programming & Development**

- Python: data analysis, visualization, geospatial workflows
- Java: object-oriented programming, algorithms, data structures
- SQL: relational databases and data querying

**Data Analysis & Visualization**

- Pandas, NumPy, Matplotlib, Plotly, Dash, Jupyter, R
- GeoPandas, GIS workflows: spatial analysis and mapping
- Geospatial datasets: MTBS, state and county shapefiles, population data
- SPSS: statistical analysis and social science research

**Research & Analytical Methods**

- Literature review: academic research and writing
- Research methods: quantitative and qualitative analysis
- Exploratory data analysis and statistical analysis
- Research data management

**Software Engineering & Systems**

- Git / GitHub: version control and collaborative development
- Virtual environments, virtual machines, and Docker / containerization

**Web Application Development**

- Dash, Leaflet: interactive web applications and mapping
- HTML, CSS, JavaScript: web development and interactive interfaces
- Flask: web applications in development stages
- Building database-backed applications

## ACADEMIC HIGHLIGHTS

**Foreign Exchange - Mälardalen University**
*Västerås, Sweden: Spring 2026*

- Exchange study and collaborative research in computer science and artificial intelligence.

**Stanford University – Summer Quarter**
*Stanford, CA: Summer 2022*

- Completed college-level coursework in data science, creative nonfiction, and dance.
- Earned a 4.0 GPA while living on campus with international peers.

## ACTIVITIES

**Grizzly Marching Band**
University of Montana: 2023 – Present

- Performed as a piccolo player in high-stakes stadium environments (20,000+ attendees), demanding real-time synchronization of skills.
- Mentored first-year members through learning drill, memorizing music, and utilizing performance anxiety techniques.

## ADDITIONAL EXPERIENCE

**Student Employee – The Lodge Dining Center**
University of Montana: August 2023 – September 2025

**Window Washing Technician**
Pane'N the Glass Window Cleaning: Summer 2024, 2025, & 2026

**Administrative Assistant**
Montana Land Homes Realty: 2022 - 2023